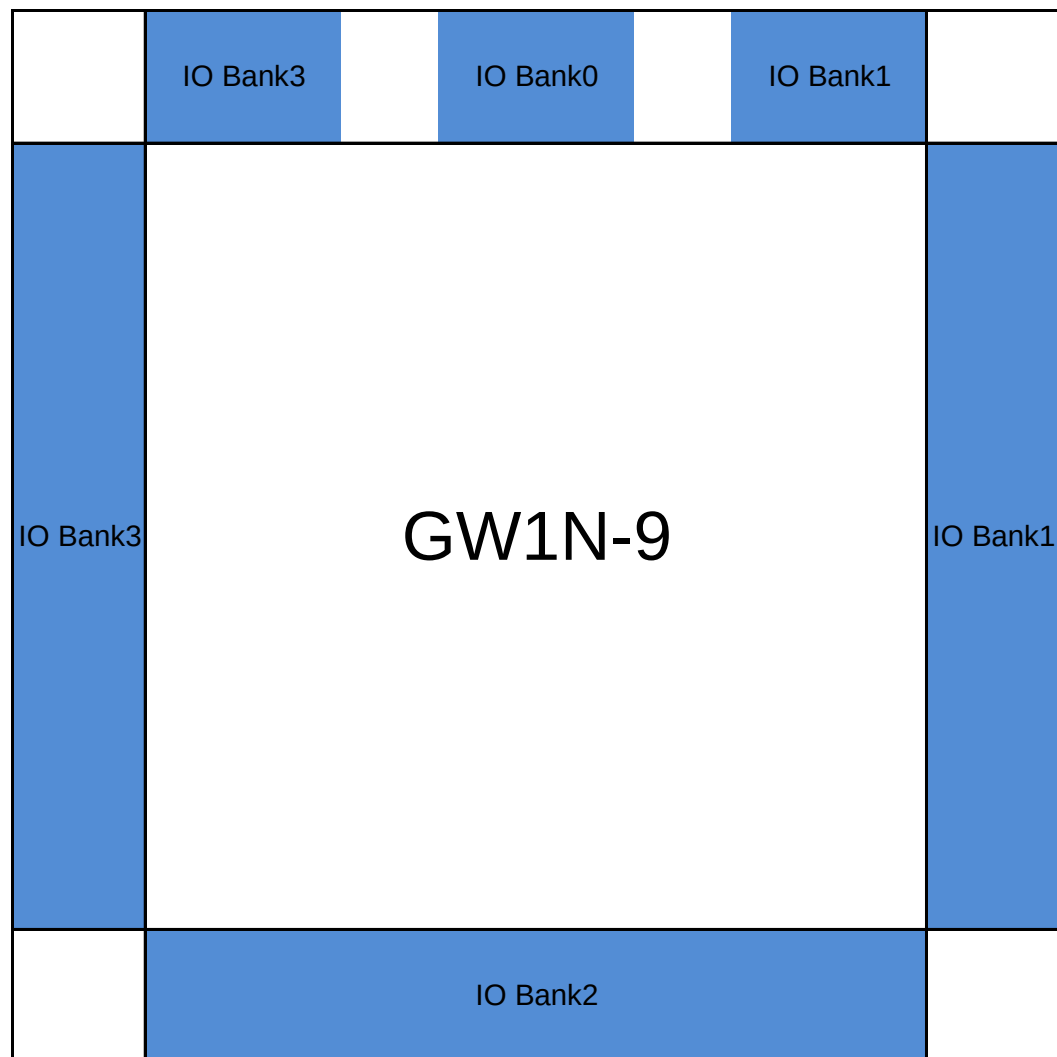


Date	Version	Description
08/23/2016	1.0E	Initial version published.
06/20/2019	1.1E	The info. of UG169 and EQ176 packages added.
03/30/2020	1.2E	The info. of CS81M package added. The info. of Power improved. The descriptions of pin MODE0/MODE1/MODE2 improved.
04/16/2020	1.3E	The info. of GW1N-6 devices removed.
07/03/2020	1.4E	The info. of MG100 and QN48F packages added.
03/17/2021	1.4.1E	The info. of True LVDS improved.
05/19/2021	1.5E	The info. of MG100T package added.
10/20/2022	1.6E	Pin definitions updated. The note in Power sheet updated.
05/04/2023	1.6.1E	The note of QN48 package in Power sheet added. The description of CLKHOLD_N pin in Pin Definitions sheet updated.
06/30/2023	1.7E	QN60 package added. The descriptions of pin definitions and info. for MODE0, MODE1, MODE2 pins optimized.
08/18/2023	1.7.1E	The info. of QN60 package in Pin List sheet and True LVDS sheet updated.
10/31/2023	1.8E	The info. of QN88F package added. The MODE2 pin of QN60 updated.
11/15/2024	1.8.1E	The description of EPAD pin in Pin Definitions sheet added.
04/30/2025	1.9E	The information of CS81M package removed.

Pin Name	I/O	Description
User I/O		
IO[End][Row/Column Number][A/B]	I/O	[End] indicates the pin location, including L(left), R(right), B(bottom), and T(top).
		[Row/Column Number] indicates the pin row/column number. If [End] is T(top) or B(bottom), the pin indicates the column number of the corresponding CFU. If [End] is L(left) or R(right), the pin indicates the row number of the corresponding CFU.
		[A/B] indicates differential signal pair information.
Multi-Function Pins		
IO[End][Row/Column Number][A/B]/MMM		/MMM represents one or more of the other functions in addition to being general purpose user I/O. When these functions are not in use, these pins can be used as user I/O.
D0	I/O	Data port D0 in CPU mode
D1	I/O	Data port D1 in CPU mode
D2	I/O	Data port D2 in CPU mode
D3	I/O	Data port D3 in CPU mode
D4	I/O	Data port D4 in CPU mode
D5	I/O	Data port D5 in CPU mode
D6	I/O	Data port D6 in CPU mode
D7	I/O	Data port D7 in CPU mode
WE_N	I	Select data input/output of D[7:0] in CPU mode.0: Write;1: Read.
DOUT	O	Data output in SERIAL mode
DIN	I, internal weak pull-up	Data input in SERIAL mode
TMS	I, internal weak pull-up	Serial mode input in JTAG mode
TCK	I	Serial clock input in JTAG mode
TDO	O	Serial data output in JTAG mode
TDI	I, internal weak pull-up	Serial data input in JTAG mode
JTAGSEL_N	I, internal weak pull-up	Reconfigure JTAG download function signal
RECONFIG_N	I, internal weak pull-up	Global reset GowinCONFIG logic signal, active low
FASTRD_N	I	Access SPI FLASH to select signal. Low, Fast Read mode; High, Read mode.
DONE ^[1]	O, internal weak pull-up	High, the programming configuration has been completed successfully; Low, the programming configuration has not been completed or failed.
	I, internal weak pull-up	When the DONE signal is low, delay the chip to activate. Activate the chip until the DONE signal is high.
READY ^[1]	I/O, internal weak pull-up	High, the device can be programmed and configured currently;
		Low, the device cannot be programmed and configured currently.

Pin Name	I/O	Description
MI	I	MI in MSPI mode
MO	O	MO in MSPI mode
MCS_N	O	Enable signal MCS_N in MSPI mode, active-low
MCLK	O	Clock output MCLK in MSPI mode, with default frequency of 2.5Mhz
SCLK	I	Clock input in SSPI, SERIAL, and CPU modes
SO	O	SO in SSPI mode
SI	I/O	SI in SSPI mode
SSPI_CS_N	I/O	Enable signal SSPI_CS_N in SSPI mode, active-low, and internal weak pull-up
CLKHOLD_N	I, internal weak pull-up	Active-high in SSPI mode; Active-low in CPU mode.
GCLKC_[x]	I	Differential input pin of GCLKT_[x], C(Comp), [x] is global clock number ^[2]
GCLKT_[x]	I	Global clock input pin, T(True), [x] is global clock number
LPLL_C_fb/RPLL_C_fb	I	Left/Right PLL feedback input pin, C(Comp)
LPLL_T_fb/RPLL_T_fb	I	Left/Right PLL feedback input pin, T(True)
LPLL_C_in/RPLL_C_in	I	Left/Right PLL clock input pin, C(Comp)
LPLL_T_in/RPLL_T_in	I	Left/Right PLL clock input pin, T(True)
MODE2	I, internal weak pull-up	GowinCONFIG modes selection pin; if this pin is marked as "VCCIO", it's internally powered; if this pin is marked as "GND", it's internally grounded.
MODE1	I, internal weak pull-up	GowinCONFIG modes selection pin; if this pin is marked as "VCCIO", it's internally powered; if this pin is marked as "GND", it's internally grounded.
MODE0	I, internal weak pull-up	GowinCONFIG modes selection pin; if this pin is marked as "VCCIO", it's internally powered; if this pin is marked as "GND", it's internally grounded.
Other Pins		
NC	NA	Reserved
VSS	NA	Ground
VCC	NA	Power supply pin of core voltage
VCCIO#	NA	Power supply pin of I/O voltage for I/O BANK#
VCCX	NA	Power supply pin of auxiliary voltage
EPAD	NA	Exposed pad. Connect to ground.
Note!		
^[1] The default state of READY/DONE is open-drain output, internal weak pull-up. DONE outputs 0 during configuration.		
^[2] When the input is single-ended, GCLKC_[x] pin is not a global clock pin.		

**Note!**

[1] Each Bank has independent reference voltage (VREF).

[2] You can select to use IOB internal VREF (equals to $0.5 * VCCIO$).

[3] You can also select to use external VREF input (use any IO pins as external VREF input).

Note!

Note! ^[1] The pin is internally grounded. ^[2] LV Version.																									
Pin Name	Function	BANK	Configuration Function	Differential Pair	LVDS	X16	QN48	CM64	QN88	LQ100	MG100	LQ144	EQ144	MG160	UG169	LQ176	EQ176	MG196	UG256	PG256	UG332	QN48F	MG100T	QN60 ^[2]	QN88F ^[2]
IOB38B	I/O	2		Comp of IOB38A	NONE	NONE													P1	P1	T3				
IOB39A	I/O	2		True of IOB39B	TRUE	x16	23	H2		49	F7	70	70	P11		83	83	N11	M3	M3	V1		F7	30	48
IOB39B	I/O	2		Comp of IOB39A	TRUE	NONE	24	G2		50	G7	71	71	N11		84	84	P11	N1	N1	U2		G7	29	47
IOB3A	I/O	2		True of IOB3B	NONE	NONE					H1										B5				
IOB3B	I/O	2		Comp of IOB3A	NONE	NONE															A5		H1		
IOB40A	I/O	2		True of IOB40B	NONE	NONE													M2	M2	T4				
IOB40B	I/O	2		Comp of IOB40A	NONE	NONE								P13					N3	N3	R5				
IOB41A	I/O	2		True of IOB41B	TRUE	x16		H1	41		K10	72	72	P12	L11	85	85	N12	R1	R1	U3		K10		
IOB41B	I/O	2		Comp of IOB41A	TRUE	NONE		G1	42		K9			N12	M11	86	86	P12	P2	P2	V2		K9		
IOB42A	I/O	2		True of IOB42B	NONE	NONE										92	92				T4	U4			
IOB42B	I/O	2		Comp of IOB42A	NONE	NONE						75	75			91	91				P4	T6			
IOB43A	I/O	2		True of IOB43B	TRUE	x16				55	J10	78	78	M14	K8	94	94	L12			T2	W2		J10	
IOB43B	I/O	2		Comp of IOB43A	TRUE	NONE		47	53			76	76	N14	J8	93	93	M12			R3	W1			
IOB44A	I/O	2		True of IOB44B	NONE	NONE								K13		97	97				R5	T7			
IOB44B	I/O	2		Comp of IOB44A	NONE	NONE								L14		96	96				P5	T8			
IOB45A	I/O	2		True of IOB45B	TRUE	x16								J13	L10	99	99	N13			T3	U5			
IOB45B	I/O	2		Comp of IOB45A	TRUE	NONE								L13	M10	98	98	P13			R4	V4			
IOB46A	I/O	2		True of IOB46B	NONE	NONE										101	101					U6			
IOB46B	I/O	2		Comp of IOB46A	NONE	NONE										100	100					V5			
IOB4A	I/O	2		True of IOB4B	TRUE	x16		19			K1	29	29	L1	M5	38	38	L2				D6		K1	
IOB4B	I/O	2		Comp of IOB4A	TRUE	NONE		20			K2	30	30	M1	M4	39	39	L1				E7		K2	
IOB5A	I/O	2		True of IOB5B	NONE	NONE															D6	C5			
IOB5B	I/O	2		Comp of IOB5A	NONE	NONE															E7	D5			
IOB6A	I/O	2		True of IOB6B	TRUE	x16				22		32	32	N1	J5	41	41	N1			A4	E6			
IOB6B	I/O	2		Comp of IOB6A	TRUE	NONE				23		34	34	P2	K5	42	42				C5	F5			
IOB7A	I/O	2		True of IOB7B	NONE	NONE															A5	B4			
IOB7B	I/O	2		Comp of IOB7A	NONE	NONE															B6	A4			
IOB8A	I/O	2		True of IOB8B	TRUE	x16	13	H8	25	27	G4	38	38	N3	N5	47	47	M2			A3	C4	13	G4	26
IOB8B	I/O	2		Comp of IOB8A	TRUE	NONE	14	G8	26	28	G3	39	39	P3	N4	48	48	M1			B4	A3	14	G3	25
IOB9A	I/O	2		True of IOB9B	NONE	NONE						40	40			49	49		D3	D3	B3				
IOB9B	I/O	2		Comp of IOB9A	NONE	NONE						41	41			50	50		D1	D1	D4				
IOL11A/TMS	I/O	3	TMS	True of IOL11B	TRUE	NONE	4	D7	5	8	E2	13	13	F1	G1	16	16	B14	B8	B8	C9	4	E2	9	5
IOL11B/TCK	I/O	3	TCK	Comp of IOL11A	TRUE	NONE	5	D6	6	9	E3	14	14	G1	G2	17	17	B13	A7	A7	A8	5	E3	8	6
IOL12A/SCLK	I/O	3	SCLK	True of IOL12B	NONE	NONE				10		15	15	F3		18	18		C10	C10	B13	6			
IOL12B/TDI	I/O	3	TDI	Comp of IOL12A	NONE	NONE	6	E7	7	11	F3	16	16	G4	F5	19	19	A13	A6	A6	B7	7	F3	10	7
IOL13A/TDO	I/O	3	TDO	True of IOL13B	TRUE	NONE	7	E6	8	12	F2	18	18	G3	F6	20	20	C14	C6	C6	C7	8	F2	12	8
IOL13B/RECONFIG_N	I/O	3	RECONFIG_N	Comp_of_IOL13A	TRUE	NONE	8		9	14	D3	20	20	H3		21	21		B10	B10	A14	9	D3		9
IOL14A/DONE	I/O	3	DONE	True of IOL14B	NONE	NONE	9		10	15		21	21	J4	H1	23	23	N14	C13	C13	B17				
IOL14B/READY	I/O	3	READY	Comp of IOL14A	NONE	NONE				16		22	22	H2					A13	A13	A13		D1		
IOL15A/GCLKT_6	I/O	3	GCLKT_6	True of IOL15B	TRUE	NONE	10	D8	11	17	F1	23	23	J2	H4	24	24	G2	C8	C8	D12	10		14	10
IOL15B/GCLKC_6	I/O	3	GCLKC_6	Comp of IOL15A	TRUE	NONE	11	E8		18	D1	24	24	J3	H5	25	25	G1	A8	A8	C12	11	F1	13	11
IOL16A	I/O	3		True of IOL16B	NONE	NONE													F8	F9	B12				
IOL16B	I/O	3		Comp of IOL16A	NONE	NONE													D9	E11	A12		D2		
IOL17A	I/O	3		True of IOL17B	TRUE	NONE								H1	G4	26	26	J2	D8		D11				
IOL17B	I/O	3		Comp of IOL17A	TRUE	NONE								K3	F4	27	27	J1	E9		C11				
IOL18A	I/O	3		True of IOL18B	NONE	NONE													B7	B9	B11				
IOL18B	I/O	3		Comp of IOL18A	NONE	NONE													C7	A10	A11				
IOL20A	I/O	3		True of IOL20B	TRUE	NONE				19	D2			J1	H3	28	28	J4	F7	F8	E10				13
IOL20B	I/O	3		Comp of IOL20A	TRUE	NONE				20	G2			K1	H2	29	29	J3	E8	D9	D10				14
IOL21A	I/O	3		True of IOL21B	NONE	NONE													C4	D8	C10				
IOL21B	I/O	3		Comp of IOL21A	NONE	NONE													B5	E9	B10		G2		
IOL22A	I/O	3		True of IOL22B	TRUE	NONE		F8	13			25	25	K2	J2	30	30	K2	E6	B7	E12				15
IOL22B	I/O	3		Comp of IOL22A	TRUE	NONE		F7	14			26	26	L2	J1	31	31	K1	D7	C7	B9		G1		16
IOL23A	I/O	3		True of IOL23B	NONE	NONE									L1				D6	F7	D9				
IOL23B	I/O	3		Comp of IOL23A	NONE	NONE									L2				E7	E8	E9				
IOL24A	I/O	3		True of IOL24B	TRUE	NONE					G1				M2	32	32		A4	C4	A9				17
IOL24B	I/O	3		Comp of IOL24A	TRUE	NONE					H2				M1	33	33		C5	B5	B8				18
IOL25A	I/O	3		True of IOL25B	NONE	NONE									N3				B3	E6	A7				

Note!

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Note!

^[1] The pin is internally grounded.

^[2] LV Version.

Pin Name	Function	BANK	Configuration Function	Differential Pair	LVDS	X16	QN48	CM64	QN88	LQ100	MG100	LQ144	EQ144	MG160	UG169	LQ176	EQ176	MG196	UG256	PG256	UG332	QN48F	MG100T	QN60 ^[2]	QN88F ^[2]	
IOT31A	I/O	0		True of IOT31B	NONE	x16					B8				B10			C8	J11		R20					
IOT31B	I/O	0		Comp of IOT31A	NONE	NONE					C8				B9			B9	L12		T20					
IOT32A	I/O	0		True of IOT32B	NONE	NONE	43			86		121	121	C8		148	148		K13	L16	U20					
IOT32B	I/O	0		Comp of IOT32A	NONE	NONE	42			85		120	120	D8		147	147		K12	L14	V20					
IOT33A	I/O	0		True of IOT33B	NONE	x16		A2			D8			B9	A8			A11	L13		T19	41	C7	49		
IOT33B	I/O	0		Comp of IOT33A	NONE	NONE		B2			D7			B10	A9			B12	M14		U19	40	B7	50		
IOT34A	I/O	0		True of IOT34B	NONE	NONE						119	119	D9		146	146	D11	D14	K13	N16					
IOT34B	I/O	0		Comp of IOT34A	NONE	NONE						118	118	C9		145	145	C11	E15	K12	N17					
IOT35A	I/O	0		True of IOT35B	NONE	x16					B9				C10			B11	K11		V19	39		48	69	
IOT35B	I/O	0		Comp of IOT35A	NONE	NONE					A10				C9			A12	M15		W19	38		47	68	
IOT36A	I/O	1		True of IOT36B	NONE	NONE								A11						K11	P16					
IOT36B	I/O	1		Comp of IOT36A	NONE	NONE								A10							L13	P17				
IOT37A	I/O	1		True of IOT37B	NONE	NONE	41		77	84		117	117			144	144		N15	M14	T17					
IOT37B	I/O	1		Comp of IOT37A	NONE	NONE	40		76	83		116	116			143	143		P16	M15	R17					
IOT38A	I/O	1		True of IOT38B	NONE	NONE			75					B11		142	142		N16	D14	W18					
IOT38B	I/O	1		Comp of IOT38A	NONE	NONE			74					B12		141	141		N14	E15	Y18					
IOT39A	I/O	1		True of IOT39B	NONE	x16		A1	73	82		115	115			140	140		P15	N15	V17		B8			
IOT39B	I/O	1		Comp of IOT39A	NONE	NONE		B1	72	81		114	114			139	139		R16	P16	U17		C8			
IOT3A	I/O	3		True of IOT3B	NONE	NONE													B14		D16					
IOT3B	I/O	3		Comp of IOT3A	NONE	NONE													A15		E15					
IOT40A	I/O	1		True of IOT40B	NONE	NONE															Y19					
IOT40B	I/O	1		Comp of IOT40A	NONE	NONE															W20					
IOT41A	I/O	1		True of IOT41B	NONE	x16			71	80		113	113			138	138			N16	U16		D7			
IOT41B	I/O	1		Comp of IOT41A	NONE	NONE			70	79		112	112			137	137			N14	R16		D8			
IOT42A	I/O	1		True of IOT42B	NONE	NONE	39		69	78		111	111	A12		136	136			P15	W16					
IOT42B	I/O	1		Comp of IOT42A	NONE	NONE	38		68	77		110	110	A13		135	135			R16	V16					
IOT43A	I/O	1		True of IOT43B	NONE	x16															Y16					
IOT43B	I/O	1		Comp of IOT43A	NONE	NONE															W15					
IOT44A	I/O	1		True of IOT44B	NONE	NONE								D10							W17					
IOT44B	I/O	1		Comp of IOT44A	NONE	NONE								C10							Y17					
IOT45A	I/O	1		True of IOT45B	NONE	x16													R12		T15		B9			
IOT45B	I/O	1		Comp of IOT45A	NONE	NONE													P13		U15		A10			
IOT46A	I/O	1		True of IOT46B	NONE	NONE													R11	R12	V15					
IOT46B	I/O	1		Comp of IOT46A	NONE	NONE													T12	P13	Y15					
IOT4A	I/O	3		True of IOT4B	NONE	x16															C19					
IOT4B	I/O	3		Comp of IOT4A	NONE	NONE															B20					
IOT5A/MODE0	I/O	3	MODE0	True_of_IOT5B	NONE	NONE	GND ^[1]	GND ^[1]	88	100	GND ^[1]	144	144	B1	GND ^[1]	174	174	GND ^[1]	GND ^[1]	M16	C17	48	GND ^[1]	GND ^[1]	GND ^[1]	
IOT5B/MODE2	I/O	3	MODE2	Comp_of_IOT5A	NONE	NONE	48	GND ^[1]	GND ^[1]	GND ^[1]	GND ^[1]	GND ^[1]	GND ^[1]	A2	GND ^[1]	173	173	GND ^[1]	GND ^[1]	C15	D17	GND ^[1]	GND ^[1]	60	GND ^[1]	
IOT6A	I/O	3		True of IOT6B	NONE	x16															F16					
IOT6B/MODE1	I/O	3	MODE1	Comp_of_IOT6A	NONE	NONE	48	GND ^[1]	87	GND ^[1]	D4	143	143	A3	GND ^[1]	172	172	GND ^[1]	GND ^[1]	B16	E17	GND ^[1]	D4	60	87	
IOT7A	I/O	3		True of IOT7B	NONE	NONE															D18					
IOT7B	I/O	3		Comp of IOT7A	NONE	NONE															B19					
IOT8A	I/O	3		True of IOT8B	NONE	x16		A8	86	99		142	142	B3		171	171				D19		A3			
IOT8B	I/O	3		Comp of IOT8A	NONE	NONE		B8	85	98		141	141	B4		170	170				C20		A4			
IOT9A	I/O	3		True of IOT9B	NONE	NONE														D16	E18					
IOT9B	I/O	3		Comp of IOT9A	NONE	NONE														E14	F18					
VCC	Power	N/A					12	E5	22	25	J2	36	36		F7	44	44						12	J2	31	45
VCC	Power	N/A					37	D4	66	75	A8	108	108		G6	132	132						37	A8	45	22
VCC	Power	N/A							1	1	A2	1	1		G8	1	1							A2	1	1
VCC	Power	N/A							45	51		73	73		H7	89	89								16	66
VCC	Power	N/A												A1					E10	A1	A1	J10				
VCC	Power	N/A												A14					E5	A16	A16	J11				
VCC	Power	N/A												P1					E6	G7	G7	K9				
VCC	Power	N/A												P14					E9	G10	G10	K12				
VCC	Power	N/A																	F10	K7	K7	L9				
VCC	Power	N/A																	F5	K10	K10	L12				
VCC	Power	N/A																	F6	T1	T1	M10				
VCC	Power	N/A																	F9	T16	T16	M11				

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Note!
^[1] The pin is internally grounded.
^[2] LV Version.

Pin Name	Function	BANK	Configuration Function	Differential Pair	LVDS	X16	QN48	CM64	QN88	LQ100	MG100	LQ144	EQ144	MG160	UG169	LQ176	EQ176	MG196	UG256	PG256	UG332	QN48F	MG100T	QN60 ^[2]	QN88F ^[2]
NC	N/A	N/A																			G1				
NC	N/A	N/A																			H3				
NC	N/A	N/A																			N18				
NC	N/A	N/A																			P20				

Note! ^[1] LV Version.																									
Pin Name	Function	BANK	Configuration Function	Differential Pair	LVDS	X16	QN48	CM64	QN88	LQ100	MG100	LQ144	EQ144	MG160	UG169	LQ176	EQ176	MG196	UG256	PG256	UG332	QN48F	MG100T	QN60 ^[1]	QN88F ^[1]
BANK3 True LVDS Pair																									
IOL11A/TMS	I/O	3	TMS	True of IOL11B	TRUE	NONE	4	D7	5	8	E2	13	13	F1	G1	16	16	B14	B8	B8	C9	4	E2	9	5
IOL11B/TCK	I/O	3	TCK	Comp of IOL11A	TRUE	NONE	5	D6	6	9	E3	14	14	G1	G2	17	17	B13	A7	A7	A8	5	E3	8	6
IOL13A/TDO	I/O	3	TDO	True of IOL13B	TRUE	NONE	7	E6	8	12	F2	18	18	G3	F6	20	20	C14	C6	C6	C7	8	F2	12	8
IOL13B/RECONFIG_N	I/O	3	RECONFIG_N	Comp of IOL13A	TRUE	NONE	8		9	14	D3	20	20	H3		21	21		B10	B10	A14	9	D3		9
IOL15A/GCLKT_6	I/O	3	GCLKT_6	True of IOL15B	TRUE	NONE	10	D8		17	F1	23	23	J2	H4	24	24	G2	C8	C8	D12	10		14	10
IOL15B/GCLKC_6	I/O	3	GCLKC_6	Comp of IOL15A	TRUE	NONE	11	E8		18	D1	24	24	J3	H5	25	25	G1	A8	A8	C12	11	F1	13	11
IOL17A	I/O	3		True of IOL17B	TRUE	NONE								H1	G4	26	26	J2	D8		D11				
IOL17B	I/O	3		Comp of IOL17A	TRUE	NONE								K3	F4	27	27	J1	E9		C11				
IOL20A	I/O	3		True of IOL20B	TRUE	NONE				19	D2			J1	H3	28	28	J4	F7	F8	E10				13
IOL20B	I/O	3		Comp of IOL20A	TRUE	NONE				20	G2			K1	H2	29	29	J3	E8	D9	D10				14
IOL22A	I/O	3		True of IOL22B	TRUE	NONE		F8	13			25	25	K2	J2	30	30	K2	E6	B7	E12				15
IOL22B	I/O	3		Comp of IOL22A	TRUE	NONE		F7	14			26	26	L2	J1	31	31	K1	D7	C7	B9				16
IOL24A	I/O	3		True of IOL24B	TRUE	NONE					G1				M2	32	32		A4	C4	A9				17
IOL24B	I/O	3		Comp of IOL24A	TRUE	NONE					H2				M1	33	33		C5	B5	B8				18
IOL26A	I/O	3		True of IOL26B	TRUE	NONE			15			27	27		L3				A5		D8				19
IOL26B	I/O	3		Comp of IOL26A	TRUE	NONE			16			28	28		M3				B6		E8				20
IOL2A	I/O	3		True of IOL2B	TRUE	NONE								C1	C2	3	3		B11	B14	A18				88
IOL2B	I/O	3		Comp of IOL2A	TRUE	NONE								D2	D1	4	4		A12	A15	A17				3
IOL4A	I/O	3		True of IOL4B	TRUE	NONE								D1	E4	7	7	E2	B13	B11	C16				
IOL4B	I/O	3		Comp of IOL4A	TRUE	NONE								E1	E3	8	8	E1	A14	A12	B16				
IOL6A/LPLL T_fb	I/O	3	LPLL T_fb	True of IOL6B	TRUE	NONE					B2			E4	B1	11	11	F4	D10	B13	A16				
IOL6B/LPLL C_fb	I/O	3	LPLL C_fb	Comp of IOL6A	TRUE	NONE					C1	6	6	F4	C1	12	12	F3	E10	A14	A15				
IOL8A	I/O	3		True of IOL8B	TRUE	NONE						9	9	H4	E1			F2	A9	D10	B15				
IOL8B	I/O	3		Comp of IOL8A	TRUE	NONE						10	10	K4	F1			F1	B9	E10	C14				
BANK2 True LVDS Pair																									
IOB11A	I/O	2		True of IOB11B	TRUE	x16			27	29	J3	42	42	L5	M7	51	51	N2	B1	B1	B1		J3	18	27
IOB11B	I/O	2		Comp of IOB11A	TRUE	NONE			28	30	H3	43	43	M5	N6	52	52	P2	C2	C2	C2		H3	19	28
IOB13A	I/O	2		True of IOB13B	TRUE	x16		H7	29	31	E4	46	46	N4	N8	53	53		E2	E2	G5		E4	20	29
IOB13B	I/O	2		Comp of IOB13A	TRUE	NONE		G7	30	32	F4	47	47	P4	N7	54	54		E3	E3	H5		F4	21	30
IOB15A	I/O	2		True of IOB15B	TRUE	x16				33	K3	48	48	N5	K6	55	55	N3	F4	F4	G4		K3	22	31
IOB15B	I/O	2		Comp of IOB15A	TRUE	NONE				34	K4	49	49	P5	J6	56	56	P3	G6	G6	F3		K4	23	32
IOB17A	I/O	2		True of IOB17B	TRUE	x16	15	H6	31	35	J4	50	50	L6	M9	57	57		G5	G5	G3	15	J4		33
IOB17B	I/O	2		Comp of IOB17A	TRUE	NONE	16	G6	32	36	H4	51	51	M6	M8	58	58		G4	G4	H4	16	H4		34
IOB21A	I/O	2		True of IOB21B	TRUE	x16					K5			L7	K7	61	61	L4			H2		K5		35
IOB21B	I/O	2		Comp of IOB21A	TRUE	NONE					K6			M7	J7	62	62	M4			H1		K6		36
IOB23A	I/O	2		True of IOB23B	TRUE	x16		H5	33		H5	52	52	N6	N12	63	63	N4	F5	F5	K4		H5		37
IOB23B	I/O	2		Comp of IOB23A	TRUE	NONE		G5	34		G5	54	54	P6	N11	64	64	P4	H6	H6	K3		G5		38
IOB25A	I/O	2		True of IOB25B	TRUE	x16								L9		67	67	N5			K1				40
IOB25B	I/O	2		Comp of IOB25A	TRUE	NONE								M9		68	68	P5			L1				39
IOB27A	I/O	2		True of IOB27B	TRUE	x16	17	H4						N8		69	69	N6	H4	H4	L2	17		25	
IOB27B	I/O	2		Comp of IOB27A	TRUE	NONE	18	G4						P8		70	70	P6	J6	J6	M1	18		26	
IOB29A/GCLKT_4	I/O	2	GCLKT_4	True of IOB29B	TRUE	x16	19	F5	35	41	J6	58	58	L8	M12	71	71	N8	L2	L2	M5	19	J6	28	42
IOB29B/GCLKC_4	I/O	2	GCLKC_4	Comp of IOB29A	TRUE	NONE	20	F4	36	42	H6	59	59	M8	M13	72	72	P8	M1	M1	M4	20	H6	27	41
IOB2A	I/O	2		True of IOB2B	TRUE	x16			17						L4	36	36				C6				
IOB2B	I/O	2		Comp of IOB2A	TRUE	NONE			18						L5	37	37				D7				
IOB31A	I/O	2		True of IOB31B	TRUE	x16			37	43	K7	62	62	N9		73	73	L8	J2	J2	N3	21	K7		
IOB31B	I/O	2		Comp of IOB31A	TRUE	NONE			38	44	K8	63	63	P9		74	74	M8	K1	K1	N4	22	K8		
IOB33A	I/O	2		True of IOB33B	TRUE	x16		H3	39	45	J7	64	64	L10		75	75	N9	K3	K3	P2		J7		
IOB33B	I/O	2		Comp of IOB33A	TRUE	NONE		G3	40		H7	65	65	M10		76	76	P9	K2	K2	P3		H7		
IOB35A	I/O	2		True of IOB35B	TRUE	x16	21			47	F6	66	66	N10	N9	79	79	N10	L1	L1	P4	24	F6		
IOB35B	I/O	2		Comp of IOB35A	TRUE	NONE	22			48	G6	67	67	P10	N10	80	80	P10	L3	L3	R3	23	G6		
IOB39A	I/O	2		True of IOB39B	TRUE	x16	23	H2		49	F7	70	70	P11		83	83	N11	M3	M3	V1		F7	30	48
IOB39B	I/O	2		Comp of IOB39A	TRUE	NONE	24	G2		50	G7	71	71	N11		84	84	P11	N1	N1	U2		G7	29	47
IOB41A	I/O	2		True of IOB41B	TRUE	x16		H1	41		K10	72	72	P12	L11	85	85	N12	R1	R1	U3		K10		

Note! [1] LV Version.																									
Pin Name	Function	BANK	Configuration Function	Differential Pair	LVDS	X16	QN48	CM64	QN88	LQ100	MG100	LQ144	EQ144	MG160	UG169	LQ176	EQ176	MG196	UG256	PG256	UG332	QN48F	MG100T	QN60 ^[1]	QN88F ^[1]
IOB41B	I/O	2		Comp_of_IOB41A	TRUE	NONE		G1	42		K9			N12	M11	86	86	P12	P2	P2	V2		K9		
IOB43A	I/O	2		True_of_IOB43B	TRUE	x16				55		78	78	M14	K8	94	94	L12		T2	W2				
IOB43B	I/O	2		Comp_of_IOB43A	TRUE	NONE				53		76	76	N14	J8	93	93	M12		R3	W1				
IOB45A	I/O	2		True_of_IOB45B	TRUE	x16								J13	L10	99	99	N13		T3	U5				
IOB45B	I/O	2		Comp_of_IOB45A	TRUE	NONE								L13	M10	98	98	P13		R4	V4				
IOB4A	I/O	2		True_of_IOB4B	TRUE	x16			19		K1	29	29	L1	M5	38	38	L2			D6		K1		
IOB4B	I/O	2		Comp_of_IOB4A	TRUE	NONE			20		K2	30	30	M1	M4	39	39	L1			E7		K2		
IOB6A	I/O	2		True_of_IOB6B	TRUE	x16				22		32	32	N1	J5	41	41			A4	E6				
IOB6B	I/O	2		Comp_of_IOB6A	TRUE	NONE				23		34	34	P2	K5	42	42			C5	F5				
IOB8A	I/O	2		True_of_IOB8B	TRUE	x16	13	H8	25	27	G4	38	38	N3	N5	47	47	M2		A3	C4	13	G4		26
IOB8B	I/O	2		Comp_of_IOB8A	TRUE	NONE	14	G8	26	28	G3	39	39	P3	N4	48	48	M1		B4	A3	14	G3		25
BANK1 True LVDS Pair																									
IOR11A/MI/D7	I/O	1	MI/D7	True_of_IOR11B	TRUE	NONE	34		62	68	G9	96	96	F14	H9	122	122		M9	P10	W13			42	61
IOR11B/MI/D6	I/O	1	MI/D6	Comp_of_IOR11A	TRUE	NONE	33		61	67	F9	95	95	G14	H8	121	121		L10	R10	Y14			41	60
IOR13A/FASTRD_N/D3	I/O	1	FASTRD_N/D3	True_of_IOR13B	TRUE	NONE				64		92	92	G11	G13	118	118	J13	M8	R9	T11				
IOR13B/SI/D2	I/O	1	SI/D2	Comp_of_IOR13A	TRUE	NONE				62		90	90	G13	G12	117	117	J14	N9	T10	U11				
IOR15A/DIN/CLKH_OLD_N	I/O	1	DIN/CLKHOLD_N	True_of_IOR15B	TRUE	NONE			54	59		86	86	H13	J13	113	113	H13	P8	T9	Y8				
IOR15B/DOUT/WE_N	I/O	1	DOUT/WE_N	Comp_of_IOR15A	TRUE	NONE			53	58		85	85	H12	H13	112	112	H14	T8	P9	W9				
IOR17A/GCLKT_3	I/O	1	GCLKT_3	True_of_IOR17B	TRUE	NONE	30	E3	52	57	F8	84	84	J14	G9	111	111	H11	T7	T7	V9	30	F8		56
IOR17B/GCLKC_3	I/O	1	GCLKC_3	Comp_of_IOR17A	TRUE	NONE	29	D3	51	56	G8	83	83	K14	G10			H12	R8	R8	W8	29	G8		55
IOR20A	I/O	1		True_of_IOR20B	TRUE	NONE					H8				J9			J11	R7	P8	V8				54
IOR20B	I/O	1		Comp_of_IOR20A	TRUE	NONE					G10				H10			J12	P7	T8	U9				53
IOR22A	I/O	1		True_of_IOR22B	TRUE	NONE		E1				82	82	E11	K12	109	109	K13	P6	M7	Y5				52
IOR22B	I/O	1		Comp_of_IOR22A	TRUE	NONE		E2	50			81	81	F11	J12	108	108	K14	T6	N7	Y4				51
IOR24A	I/O	1		True_of_IOR24B	TRUE	NONE	28	F1	49		H9	80	80	J12	K11	107	107	L13	T3	N6	W5	28		37	50
IOR24B	I/O	1		Comp_of_IOR24A	TRUE	NONE	27	F2	48		H10	79	79	H11	L12	106	106	L14	R4	L7	V6	27		36	49
IOR26A	I/O	1		True_of_IOR26B	TRUE	NONE								K12	K10	105	105	M13	T2		Y3			35	
IOR26B	I/O	1		Comp_of_IOR26A	TRUE	NONE								K11	J10	104	104	M14	R3		W4			34	
IOR2A	I/O	1		True_of_IOR2B	TRUE	NONE					C10				D12			C12	R13						
IOR2B	I/O	1		Comp_of_IOR2A	TRUE	NONE					B10				D11			C13	T14						
IOR4A	I/O	1		True_of_IOR4B	TRUE	NONE									B13			E13	P12	R13	T14				
IOR4B	I/O	1		Comp_of_IOR4A	TRUE	NONE									A12			E14	T13	T14	T13				
IOR6A/RPLL_T_fb	I/O	1	RPLL_T_fb	True_of_IOR6B	TRUE	NONE		D1				102	102	D14	B12	127	127	F11	T11	P12	Y13				
IOR6B/RPLL_C_fb	I/O	1	RPLL_C_fb	Comp_of_IOR6A	TRUE	NONE		D2				101	101	E14	B11	126	126	F12	P11	T13	Y12				
IOR8A	I/O	1		True_of_IOR8B	TRUE	NONE						100	100	D13	F9			G13	P10	T11	U12				63
IOR8B	I/O	1		Comp_of_IOR8A	TRUE	NONE						99	99	E13	F10			G14	R10	P11	V11				62

Note!

VCCX should be greater than or equal to VCCIO.

Recommended Operating Conditions of QN48 Package in GW1N-9

Name	Description	Min.	Max.
VCC	LV: Core voltage	1.14V	1.26V
	UV: Core voltage	1.71V	3.6V
VCCIO1/VCCIO2	VCCIO1 and VCCIO2 are internally short-circuited.	LV: I/O Bank voltage	1.14V
		UV: I/O Bank voltage	1.14V
		When MIPI output is used in BANK2, VCCIO2 should provide 1.2V voltage.	1.14V
VCCIO0/VCCIO3	VCCIO0 and VCCIO3 are internally short-circuited.	LV: I/O Bank voltage	1.14V
		UV: I/O Bank voltage	1.14V
VCCX	Auxiliary voltage	2.375V	3.6V

Note !

It is highly recommended that the EPAD connect to GND, but not a requirement.

Recommended Operating Conditions of CM64 Package in GW1N-9

Name	Description	Min.	Max.
VCC	LV: Core voltage	1.14V	1.26V
	UV: Core voltage	1.71V	3.6V
VCCIO0/VCCIO2	VCCIO0 and VCCIO2 are internally short-circuited.	LV: I/O Bank voltage	1.14V
		UV: I/O Bank voltage	1.14V
		When MIPI output is used in BANK2, VCCIO2 should provide 1.2V voltage.	1.14V
VCCIO1/VCCIO3	VCCIO1 and VCCIO3 are internally short-circuited.	LV: I/O Bank voltage	1.14V
		UV: I/O Bank voltage	1.14V
VCCX	Auxiliary voltage	2.375V	3.6V

Recommended Operating Conditions of MG100/MG100T Packages in GW1N-9

Name	Description	Min.	Max.
VCC	Core voltage	1.14V	1.26V
VCCIO0, VCCIO1, VCCIO2, VCCIO3	I/O Bank voltage	1.14V	3.6V
	When MIPI output is used in BANK2, VCCIO2 should provide 1.2V voltage.	1.14V	1.26V
VCCX	Auxiliary voltage	2.375V	3.6V

Note!

VCCX should be greater than or equal to VCCIO.

Recommended Operating Conditions of QN48F Package in GW1N-9

Name	Description	Min.	Max.
VCC	Core voltage	1.14V	1.26V
VCCIO0, VCCIO3	I/O Bank voltage	1.14V	3.6V
VCCIO1/VCCIO2	VCCIO1 and VCCIO2 are internally short-circuited.	I/O Bank voltage	1.14V
	When MIPI output is used in BANK2, VCCIO2 should provide 1.2V voltage.	1.14V	3.6V
VCCX	Auxiliary voltage	1.14V	1.26V
		2.375V	3.6V

Note !

It is highly recommended that the EPAD connect to GND, but not a requirement.

Recommended Operating Conditions of QN60 Package in GW1N-9

Name	Description	Min.	Max.
VCC	LV: Core voltage	1.14V	1.26V
VCCIO0, VCCIO1, VCCIO2, VCCIO3	LV: I/O Bank voltage	1.14V	3.6V
	When MIPI output is used in BANK2, VCCIO2 should provide 1.2V voltage.	1.14V	1.26V
VCCX	Auxiliary voltage	2.375V	3.6V

Note !

It is highly recommended that the EPAD connect to GND, but not a requirement.

Recommended Operating Conditions of QN88F Package in GW1N-9

Name	Description	Min.	Max.
VCC	LV: Core voltage	1.14V	1.26V
VCCIO1, VCCIO2	LV: I/O Bank voltage	1.14V	3.6V
VCCIO3	When MIPI output is used in BANK2, VCCIO2 should provide 1.2V voltage.	1.14V	1.26V
VCCIO0/VCCX	Auxiliary voltage VCCX and VCCIO0 are internally short-circuited.	2.375V	3.6V

Note !

It is highly recommended that the EPAD connect to GND, but not a requirement.

Note!

VCCX should be greater than or equal to VCCIO.

Recommended Operating Conditions of Other Packages in GW1N-9

Name	Description	Min.	Max.
VCC	LV: Core voltage	1.14V	1.26V
	UV: Core voltage	1.71V	3.6V
VCCIO0, VCCIO1, VCCIO2, VCCIO3	LV: I/O Bank voltage	1.14V	3.6V
	UV: I/O Bank voltage	1.14V	3.6V
	When MIPI output is used in BANK2, VCCIO2 should provide 1.2V voltage.	1.14V	1.26V
VCCX	Auxiliary voltage	2.375V	3.6V